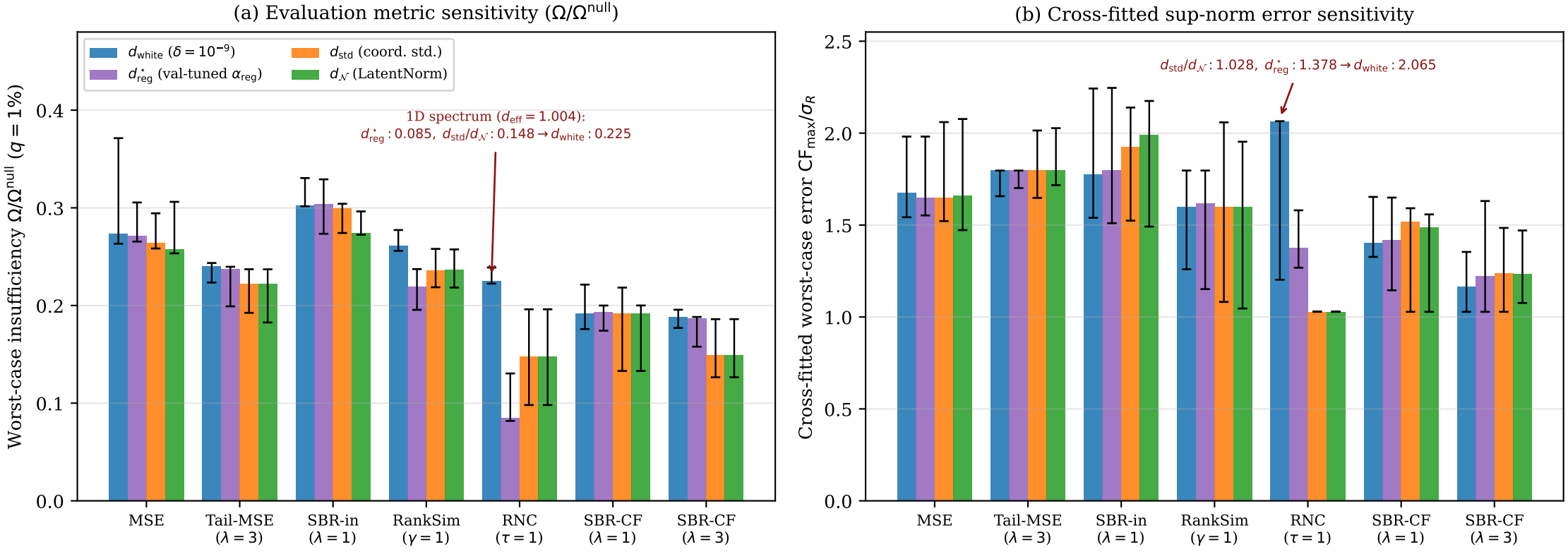
(a) Evaluation metric sensitivity ($\Omega/\Omega^{\text{null}}$)
d_{white} ($\delta = 10^{-9}$)
$d_{\text{reg}}^{\star}$ (val-tuned α_{reg})
d_{std} (coord. std.)
$d_{\mathcal{N}}$ (LatentNorm)
1D spectrum ($d_{\text{eff}} = 1.004$):
$d_{\text{reg}}^{\star}: 0.085$, $d_{\text{std}}/d_{\mathcal{N}}: 0.148 \to d_{\text{white}}: 0.225$
Worst-case insufficiency $\Omega/\Omega^{\text{null}}$ ($q = 1\%$)
0.0
0.1
0.2
0.3
0.4
MSE
Tail-MSE ($\lambda = 3$)
SBR-in ($\lambda = 1$)
RankSim ($\gamma = 1$)
RNC ($\tau = 1$)
SBR-CF ($\lambda = 1$)
SBR-CF ($\lambda = 3$)
(b) Cross-fitted sup-norm error sensitivity
$d_{\text{std}}/d_{\mathcal{N}}: 1.028$, $d_{\text{reg}}^{\star}: 1.378 \to d_{\text{white}}: 2.065$
Cross-fitted worst-case error $\text{CF}_{\text{max}}/\sigma_R$
0.0
0.5
1.0
1.5
2.0
2.5
MSE
Tail-MSE ($\lambda = 3$)
SBR-in ($\lambda = 1$)
RankSim ($\gamma = 1$)
RNC ($\tau = 1$)
SBR-CF ($\lambda = 1$)
SBR-CF ($\lambda = 3$)